Unit 5 e-Portfolio Activity: Jaccard Distance/Dissimilarity Calculations

Introduction

The brief asks for three Jaccard distances from a table of pathological test results. The harder question is what a zero means, because Jaccard ignores features on which both people score zero.

Data preparation

Following the brief, Y and P became 1, N and A became 0, and Name and Gender were left out. For each pair, f11 counts features positive for both people, f10 and f01 those positive for one only, and f00 those negative for both (University of Essex Online, 2026). Jaccard distance, one minus the Jaccard coefficient, is (f10 + f01) / (f11 + f10 + f01), so f00 plays no part (Han, Kamber and Pei, 2012).

Results

The distances are 1/3 ≈ 0.33 for (Jack, Mary), 2/3 ≈ 0.67 for (Jack, Jim) and 3/4 = 0.75 for (Jim, Mary), as in the brief (Table 2).

What the answer depends on

The brief does not define A; it may mean ‘not tested’. Excluding, for each pair, any feature where either person has A leaves all three Jaccard distances unchanged, but only because every A in this table meets a 0. An A opposite a P would count as a mismatch under the brief’s zero coding but would drop out under this exclusion, so the two policies could then disagree. Simple-matching distance, which counts shared zeros (University of Essex Online, 2026), changes for every pair (Table 3). Where zeros are real negatives, counting them may help: measures that do so performed better on Indonesian and Japanese herbal medicine data (Wijaya et al., 2016).

As a sensitivity assumption, not a clinical choice, adding Gender as one equally weighted symmetric feature, in the mixed form of Han, Kamber and Pei (2012), gives (f10 + f01 + g) / (f11 + f10 + f01 + 1), where g is 1 if the genders differ. (Jack, Mary) and (Jack, Jim) then tie at 0.50, so Jack's nearest match depends on whether Gender counts.

Implications

If these profiles represented identifiable real individuals, they would contain health data, which EU law protects as a special category (*Regulation (EU) 2016/679*, 2016, art. 9). Removing the names would not by itself make the table anonymous: Mary is the only F, so she could be singled out or linked if other identifying information were available (*Regulation (EU) 2016/679*, 2016, recital 26). Leaving Gender out does not guarantee fairness either, because other data can act as proxies, as occupation can for gender (Centre for Data Ethics and Innovation, 2019).

Conclusion

These are worked sensitivity checks on three people, not clinical validation. Before using such distances for clustering, I would confirm what A means and record which attributes and which measure the intended use justifies.

Evidence

Table 1: The brief's table and its coding (1 = Y or P; 0 = N or A). Name and Gender are not used in the Jaccard distance.

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y (1)	N (0)	P (1)	N (0)	N (0)	A (0)
Mary	F	Y (1)	N (0)	P (1)	A (0)	P (1)	N (0)
Jim	M	Y (1)	P (1)	N (0)	N (0)	N (0)	A (0)

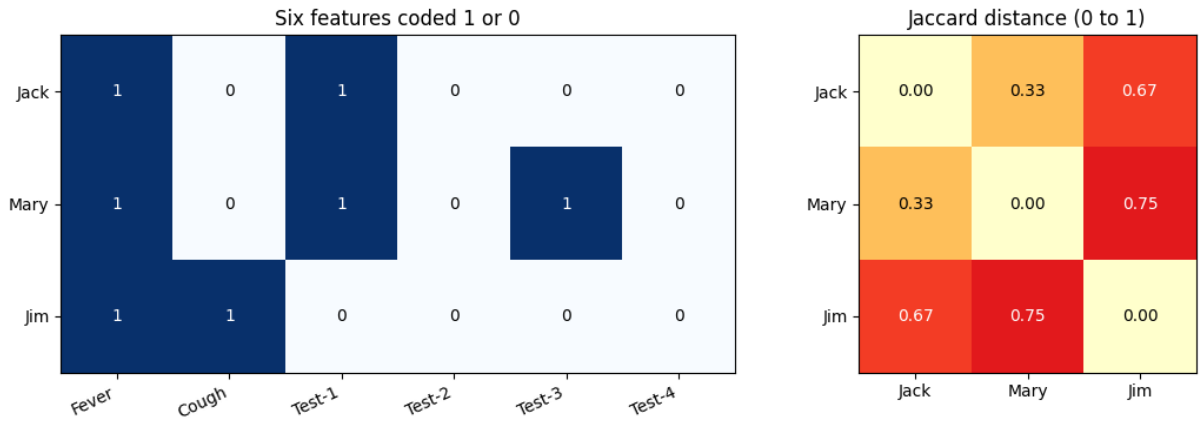
Table 2: Counts and Jaccard distance for each pair. In my notebook run, the count formula, Python sets and SciPy gave the same values.

Pair	Shared positives	f11	f10	f01	f00	Distance
(Jack, Mary)	Fever, Test-1	2	0	1	3	1/3 ≈ 0.33
(Jack, Jim)	Fever	1	1	1	3	2/3 ≈ 0.67
(Jim, Mary)	Fever	1	1	2	2	3/4 = 0.75

Table 3: Sensitivity checks. ‘A missing’ excludes, for each pair, any feature where either person has A. Simple-matching distance divides mismatches by all features compared. With Gender uses (f10 + f01 + g) / (f11 + f10 + f01 + 1), with equal weights as a sensitivity assumption.

Pair	Jaccard	Jaccard, A missing	Simple-matching distance	Simple-matching distance, A missing	With Gender
(Jack, Mary)	0.33	0.33	0.17	0.25	0.50
(Jack, Jim)	0.67	0.67	0.33	0.40	0.50
(Jim, Mary)	0.75	0.75	0.50	0.75	0.80

Figure 1: The six features coded 1 or 0 (left) and the Jaccard distance matrix (right), from my notebook run in Google Colab on 30 October 2026 at 18:44 UTC. Notebook: https://github.com/Tanky0/Tanky0.github.io/blob/main/assets/documents/ML_Unit05_Jaccard_Notebook.ipynb



References

Centre for Data Ethics and Innovation (2019) *Snapshot paper: AI and personal insurance*. Available at: https://www.gov.uk/government/publications/cdei-publishes-its-first-series-of-three-snapshot-papers-ethical-issues-in-ai/snapshot-paper-ai-and-personal-insurance (Accessed: 20 October 2026).

Han, J., Kamber, M. and Pei, J. (2012) *Data Mining: Concepts and Techniques*. 3rd edn. Waltham: Morgan Kaufmann.

Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) (2016) *Official Journal* L 119, 4 May, pp. 1–88. Available at: https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng (Accessed: 30 October 2026).

University of Essex Online (2026) 'Unit 5 Lecturecast: Clustering' [Lecturecast]. ML_PCOM7E: Machine Learning October 2026. University of Essex Online. Available at: https://www.my-course.co.uk/mod/scorm/view.php?id=1434575 (Accessed: 25 October 2026).

Wijaya, S.H. et al. (2016) 'Finding an appropriate equation to measure similarity between binary vectors: case studies on Indonesian and Japanese herbal medicines', *BMC Bioinformatics*, 17, 520. Available at: https://doi.org/10.1186/s12859-016-1392-z

AI Acknowledgement

Codex was used to assist in identifying appropriate code solutions and to proofread and refine the code I produced. Claude and Grammarly supported proofreading, language refinement, and formatting of my written text. All final decisions, analyses, interpretations, conclusions, and recommendations are my own.